

Semigroup and Groups

9.1 二元运算/Binary Operations Revisited

9.2 半群/Semigroups

9.3 乘积半群与商半群/Products and Quotients of Semigroups

9.4 群Groups

9.5 乘积群与商群Products and Quotients of Groups

9.2 Semigroups

Semigroups

- Definition (半群定义)
- Isomorphism (同构)
- Homomorphism (同态)

定义 S 上的二元运算是一个处处有定义的函数 $f: S \times S \rightarrow S$.

定义 设 S 为集合, n 为正整数, 函数

$$f: \underbrace{S \times S \times \dots \times S}_{n \text{ 个}} \rightarrow S$$

称为 S 上的 n 元运算, 简称为 n 元运算.

定义 : 非空集合 S 和 S 上若干个定义在该集合上的运算 $f_1, f_2, \dots, f_k$ 组成的系统称为一个代数系统, 简称代数, 记做 $V = (S, f_1, f_2, \dots, f_k)$

定义1：代数系统 $(S, *)$, 若 S 非空且 $*$ 是封闭的, 则称 $(S, *)$ 为广群。

定义2：代数系统 $(S, *)$, 若 S 非空且 $*$ 是封闭的, 可结合的, 则称 $(S, *)$ 为半群。

Semigroup – 半群

A *semigroup* is a *nonempty* (非空) set S together with an associative binary operation $*$ defined on S .

- Denote the semigroup by $(S, *)$ or S .
- $a * b$ is referred as the product of a and b . The semigroup $(S, *)$ is said to be *commutative* (交换半群) if $*$ is a commutative operation.

Examples

- ① $(\mathbb{Z}, +)$ is a **commutative** semigroup. (实数加法)
- ② The set $P(S)$, where S is a set, together with the operation of union is a **commutative** semigroup. (幂集 $\cup$ 运算交换半群)
- ③ $(\mathbb{Z}, -)$, The set $\mathbb{Z}$ with the binary operation of subtraction is **not a semigroup**, since subtraction is not associative. (实数减法不可结合)

Example 4

Let S be a fixed nonempty set, and let S^S be the set of all functions $f : S \rightarrow S$. If f and g are elements of S^S , we define $f * g$ as $f \circ g$, the composite function.

S 非空, $S^S = \{f \mid f: S \rightarrow S\}$, $f, g \in S^S$, 定义 $f * g$ 是函数合成运算 $f \circ g$,

运算符 $*$ 是 S^S 上的二元运算

$*$ 是可结合的.

$(S^S, *)$ 是半群, 但不是交换半群.

The semigroup S^S is not commutative.

Example 6

Let

$A = \{a_1, a_2, \dots, a_n\}$ be a nonempty set.

A^* is the set of all finite sequences of elements of A .

α and β be elements of A^* .

The catenation is a binary operation $\bullet$ on A^* .

if $\alpha = a_1 a_2 \dots a_n$ and $\beta = b_1 b_2 \dots b_k$

$$\alpha \bullet \beta = a_1 a_2 \dots a_n b_1 b_2 \dots b_k$$

if α , β , and γ are any elements of A^* ,

$$\alpha \bullet (\beta \bullet \gamma) = (\alpha \bullet \beta) \bullet \gamma.$$

$(A^*, \bullet)$ is called the *free semigroup generated by A* (由 A 生成的自由半群).

Theorem 1

If $a_1, a_2, \dots, a_n, n \geq 3$, are arbitrary elements of a semigroup, then all products of the elements $a_1, a_2, \dots, a_n$ that can be formed by inserting meaningful parentheses arbitrarily are equal.

Proof are omitted

Note

Theorem 1 shows that the products

$$((a_1 * a_2) * a_3) * a_4 ;$$

$$a_1 * (a_2 * (a_3 * a_4)) ;$$

$$(a_1 * (a_2 * a_3)) * a_4 ;$$

are all equal.

If $a_1, a_2, \dots, a_n$ are elements in a semigroup $(S, *)$, then the product can be written as

$$a_1 * a_2 * \dots * a_n$$

Identity – 单位元（幺元）

An **element** e in a semigroup $(S, *)$ is called an **identity** element if $e * a = a * e = a$ for all $a \in S$.

an identity element must be unique.

Example 8,9

the semigroup $(\mathbb{Z}, +)$.

The semigroup $(\mathbb{Z}^+, +)$

Monoid – 独异点（含幺半群）

A **monoid** is a semigroup $(S, *)$ that has an identity.

Example 10:

The semigroup $(P(S), \cup)$, 单位元为: $\emptyset$,
 $\forall A \in P(S)$.

$$\emptyset * A = \emptyset \cup A = A = A \cup \emptyset = A * \emptyset$$

Hence $P(S)$ is a monoid.

Example 11, 12, 13

Example 11: The semigroup S^S , the identity (幺元为)

I_S , * 定义为合成运算

$$I_S * f = I_S \circ f = f \circ I_S = f * I_S = f, \quad \forall f \in S^S,$$

$(S^S, *)$ is a monoid.

Example 12: The semigroup A^* , the identity (幺元为) Λ the empty sequence (空序列),

$$\alpha \cdot \Lambda = \Lambda \cdot \alpha = \alpha, \quad \forall \alpha \in A^*.$$

Example 13: The set of all relations on set A is a monoid under the operation of composition. The identity element is the equality relation.

examples

$\mathbb{Z}^+$, $a*b$ defined as $\max\{a,b\}$

$\mathbb{Z}^+$, $a*b$ defined as a

$S=\{1,2,3,6,12\}$, $a*b$ defined as $\text{GCD}(a,b)$

Subsemigroup – 子半群

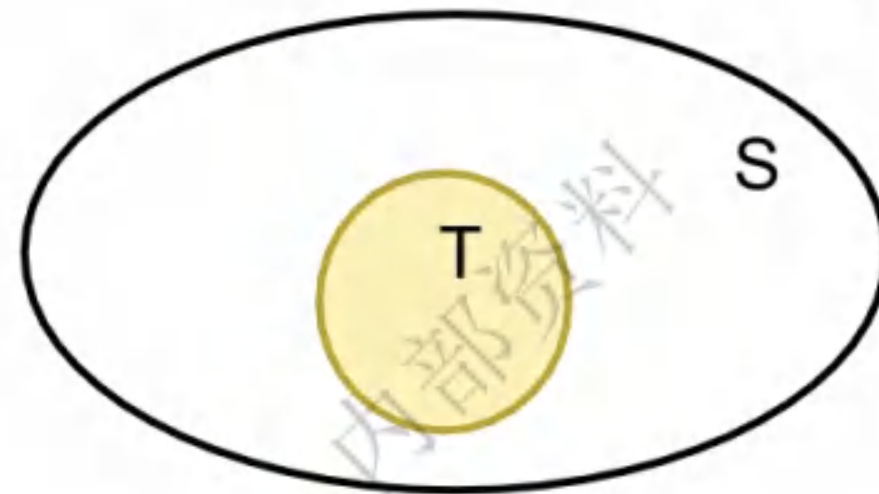
Let

$(S, *)$ be semigroup and

$$T \subseteq S$$

if T is **closed** under the operation $*$, then $(T, *)$ is called **subsemigroup** of $(S, *)$.

- 子集
- 运算封闭
- **结合性?**



Submonoid – 子独异点

Let

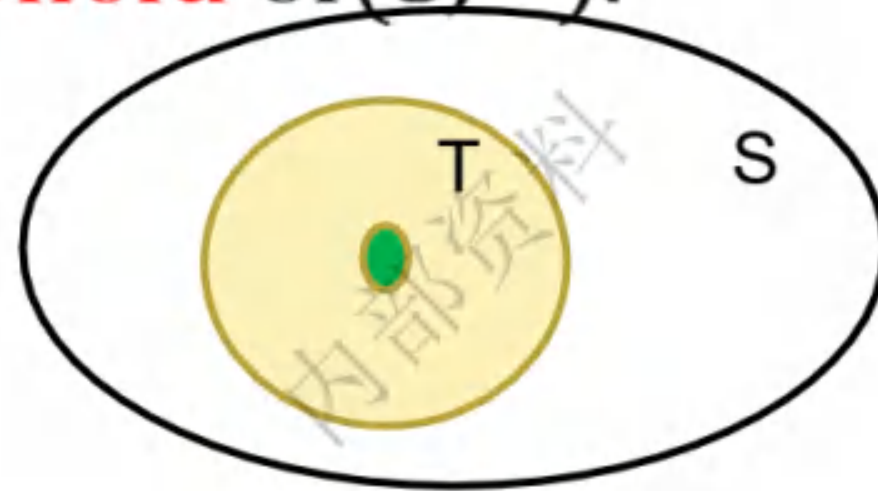
$(S, *)$ be a monoid with identity e , and

$$T \subseteq S.$$

if T is **closed** under the operation $*$ and $e \in T$, then

$(T, *)$ is called a **submonoid** of $(S, *)$.

- 子集
- 运算封闭
- 幺元属于 T



子独异点反例

半群: $T \subseteq S$, 二元运算 $*$ 为矩阵乘

$$S = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in R \right\}, S \text{ 的单位元为 } \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$T = \left\{ \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \mid a \in R \right\}, T \text{ 的单位元 } \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

所以此例中子半群 T , 不是 S 的子么半群

Note

The associative property holds in any subset of a semigroup so that a subsemigroup $(T, *)$ of a semigroup $(S, *)$ is itself a semigroup.

Similarly, a submonoid of a monoid is itself a monoid.

Examples 14, 15

Examples 14: If T is the set of all even integers, then $(T, *)$ is a subsemigroup of the monoid $(\mathbb{Z}, *)$, where $*$ is ordinary multiplication, but it is not a submonoid since the identity of $\mathbb{Z}$, the number 1, does not belong to T .

T 为偶数集合, $(T, \times)$, 是半群 $(\mathbb{Z}, \times)$ 的 **子半群**, $(\mathbb{Z}, \times)$ 为幺半群, 幺元: 1, 但 T 不是 $\mathbb{Z}$ 的 **子幺半群**.

Examples 15: If $(S, *)$ is a semigroup, then $(S, *)$ is a subsemigroup of $(S, *)$.

Similarly, let $(S, *)$ be a monoid, then $(S, *)$ is a submonoid of $(S, *)$. If $T = \{e\}$, then $(T, *)$ is also a submonoid of $(S, *)$.

Powers of a

a 的幂

Suppose

$(S, *)$ is a semigroup, $a \in S$ $n \in \mathbb{Z}^+$

Define the powers of a^n recursively as follows:

$$a^1 = a, \quad a^n = a^{n-1} * a, \quad n \geq 2.$$

$$a^0 = e \quad \text{if } (S, *) \text{ is a monoid}$$

if m and n are nonnegative integers, then

$$a^m * a^n = a^{m+n}$$

(m, n 一定为非负整数)

Example 16

-生成的半群

(a) If $(S, *)$ is a semigroup,
 $a \in S$, and $T = \{a^i \mid i \in \mathbb{Z}^+\}$,
then $(T, *)$ is a subsemigroup of $(S, *)$.

(b) If $(S, *)$ is a monoid,
 $a \in S$, and $T = \{a^i \mid i \in \mathbb{Z}^+ \text{ or } i = 0\}$,
then $(T, *)$ is a submonoid of $(S, *)$.

generated by a (由 a 生成的)

例：T为半群，任取 $a, b \in T$ ，如果 $a \neq b$ ，则 $ab \neq ba$
求：

(1) T中幂等律成立

(2) $\forall a, b \in T, aba = a$

证：(1) 反正：假设 $aa \neq a$ ，

$(aa)a \neq a(aa) \Rightarrow aaa \neq aaa$ ，矛盾

(2) 假设 $aba \neq a$ ，则

$(aba)a \neq a(aba) \Rightarrow aba \neq aba$ ，矛盾

定理：若干子半群的**非空**交集仍为子半群
若干子独异点的交集仍为子独异点。

证明思路：

(1) 设 A_1 和 A_2 是 A 的子半群，非空

则存在任意的 $a * b \in A_1$, $a * b \in A_2$

$\Rightarrow a * b \in A_1 \cap A_2$

(2) 设 A_1 和 A_2 是 A 的子独异点，则交集 $A_1 \cap A_2$
至少含有一个单位元 e

Isomorphism – 同构

Homomorphism – 同态

Isomorphism – 同构

Let $(S, *)$ and $(T, *')$ be two semigroups.

A function $f: S \rightarrow T$

is called an *isomorphism* from $(S, *)$ to $(T, *')$

if it is a one-to-one correspondence from S to T ,

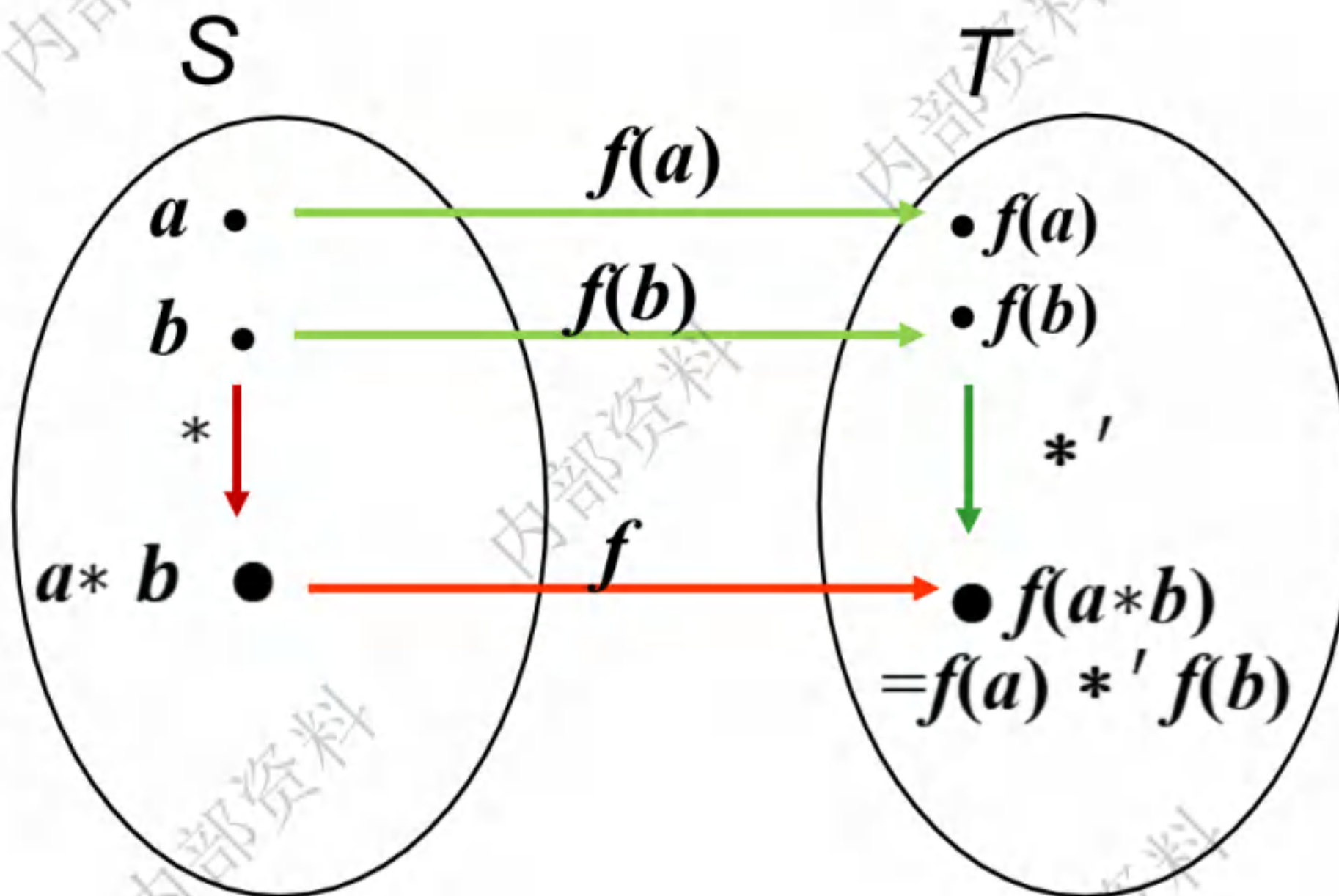
and $f(a * b) = f(a) *' f(b)$ for all a and b in S .

设 $(S, *)$ 和 $(T, *')$ 为两个半群，如果函数

$f: S \rightarrow T$ 是 S 到 T 的一一对应，并且对 S 中所有的 a 和 b 有

$$f(a * b) = f(a) *' f(b)$$

则称函数 $f: S \rightarrow T$ 是从 $(S, *)$ 到 $(T, *')$ 的一个同构



Note

If f is an isomorphism from $(S, *)$ to $(T, *')$, then, since f is a one-to-one correspondence, f^{-1} exists and is a one-to-one correspondence from T to S .

- 一一映射
- 存在反函数 f^{-1} : 从 $T \rightarrow S$ 的映射

f^{-1} is an isomorphism

Let a' and b' be any elements of T . ($a', b' \in T$)

Since f is onto, there exist ($\exists a, b \in S$)

($\forall a', b' \in T$), ($\exists a, b \in S$)

such that

$$f(a) = a' \text{ and } f(b) = b'.$$

Then $a = f^{-1}(a')$ and $b = f^{-1}(b')$.

$$f^{-1}(a' * b') = f^{-1}(f(a) * f(b))$$

$$= f^{-1}(f(a * b)) = (f^{-1} \circ f)(a * b)$$

$$= a * b = f^{-1}(a') * f^{-1}(b').$$

Hence f^{-1} is an isomorphism.

Isomorphic semigroups

同构半群

If f is an isomorphism from $(S, *)$ to $(T, *')$,
the semigroups $(S, *)$ and $(T, *')$ are said to
be *isomorphic* and write as $S \cong T$

Isomorphic?

证明同构的步骤

STEP 1: Define a function $f:S \rightarrow T$ with $\text{Dom}(f) = S$. (定义一个函数)

STEP 2: Show that f is one-to-one. (证明单射)

STEP 3: Show that f is onto. (证明满射)

STEP 4: Show that $f(a * b) = f(a) *' f(b)$ (验证定义)

Example 17

Let T be the set of all even integers.

Show that the semigroups $(\mathbb{Z}, +)$ and $(T, +)$ are isomorphic.

Example 17 - Solution

Step 1: define the function $f: \mathbb{Z} \rightarrow T$ by

$$f(a) = 2a.$$

Step 2: f is one-to-one.

Suppose that $f(a_1) = f(a_2)$.

Then $2a_1 = 2a_2$,

so $a_1 = a_2$.

Hence f is one-to-one

Example 17 - Solution

Step 3: f is onto.

Suppose that b is any even integer.

Then $a = b/2 \in \mathbb{Z}$ and

$$f(a) = f(b/2) = 2(b/2) = b,$$

so f is onto.

Step 4: $f(a+b) = 2(a+b) = 2a+2b = f(a)+f(b)$.

Hence $(\mathbb{Z}, +)$ and $(T, +)$ are isomorphic semigroups

Q.E.D.

Example 18

Let

$$S = \{a, b, c\} \text{ and } T = \{x, y, z\}.$$

It is easy to verify that S and T are semigroup.

Let

$$f(a) = y$$

$$f(b) = x$$

$$f(c) = z$$

*	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

*	x	y	z
x	z	x	y
y	x	y	z
z	y	z	x

Replacing the elements in S by their images and rearranging the table, we obtain exactly the table for T . Thus S and T are isomorphic

Theorem 2

独异点同构

Let

$(S, *)$ and $(T, *')$ be monoids with identities e and e' , respectively.

$f: S \rightarrow T$ be an isomorphism.

Then

Theorem 2 - Proof

Let b be any element of T . Since f is onto, there is an element a in S such that

$$\forall b \in T, \exists a \in S$$

$$f(a) = b.$$

Since $a = a * e$,

$$\begin{aligned} b &= f(a) = f(a * e) = f(a) *' f(e) \\ &= b *' f(e). \end{aligned} \quad (\text{右幺元})$$

Theorem 2 - Proof

Similarly, $a = e * a$,
 $b = f(e) *' b$.

Thus for any $b \in T$,

$$b = b *' f(e) = f(e) *' b, \quad (\text{左幺元})$$

which means that $f(e)$ is an identity for T .

the identity is unique,

$$f(e) = e'$$

Q.E.D.

Note

If $(S, *)$ and $(T, *')$ are semigroups such that S has an identity and T does not, it then follows from Theorem 2 that $(S, *)$ and $(T, *')$ cannot be isomorphic.

如果一个代数系统有么元，另一个没有么元
一定不同构

Example 19

Let T be the set of all even integers and let $*$ be ordinary multiplication.

the semigroups $(\mathbb{Z}, *)$ and $(T, *)$ are not isomorphic, since $\mathbb{Z}$ has an identity and T does not.

Homomorphism – 同态

- Let $(S, *)$ and $(T, *')$ be two semigroups.

An everywhere-defined function $f: S \rightarrow T$ is called a **homomorphism** from $(S, *)$ to $(T, *')$

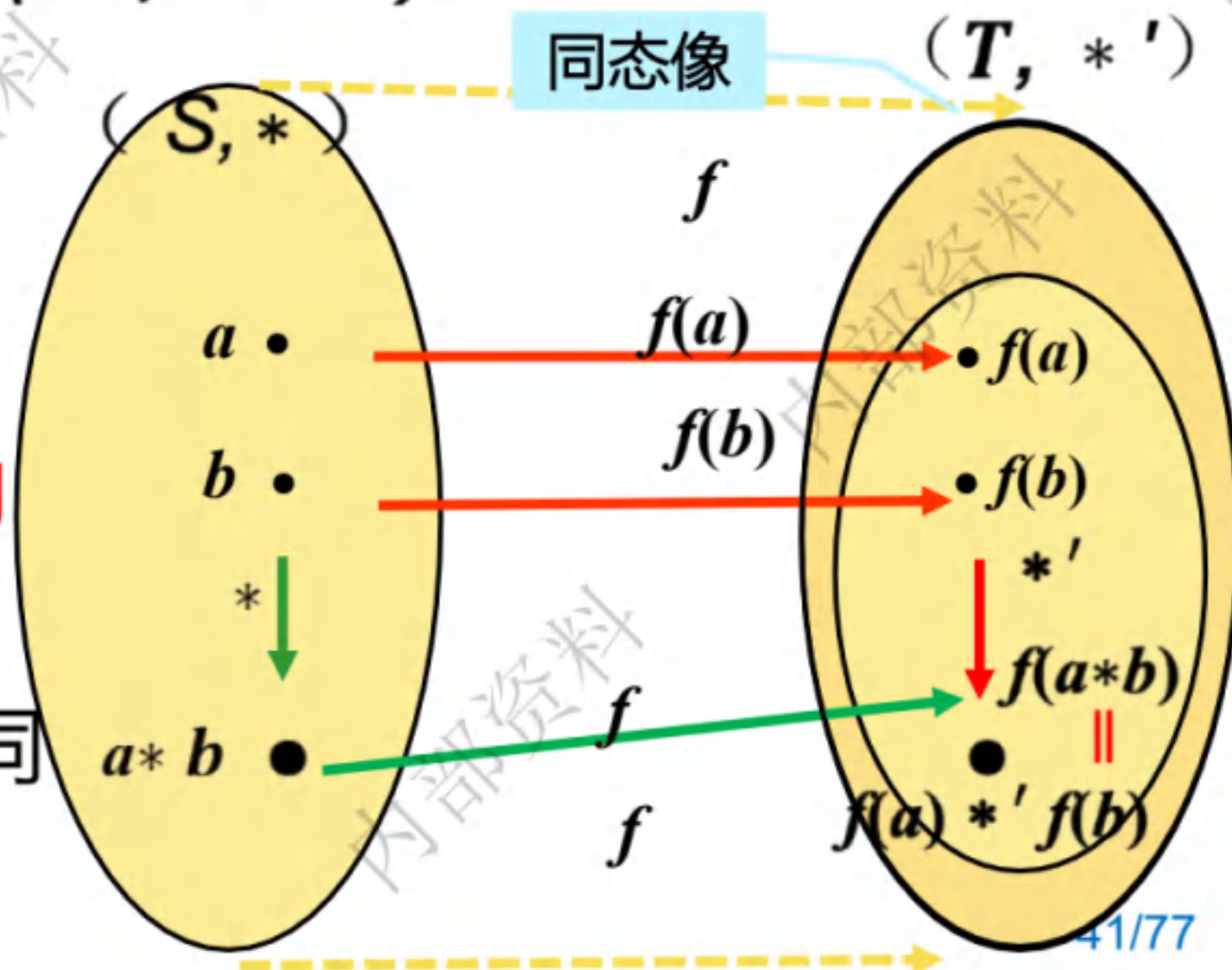
If $f(a * b) = f(a) *' f(b) \ (\forall a, b \in S)$

for all a and b in S .

- If f is also onto, we say that T is a **homomorphic image** (同态像) of S .

如果 f 是满射的, 则 T 是 S 的同态像

单同态, 自同态



Example 20

+	0	1
0	0	1
1	1	0

Let

$$A = \{0, 1\}$$

$(A^*, \bullet)$, $\bullet$ is the catenation operation (字符连接半群)

$(A, +)$, $+$ is defined by the table above (A 为 $+$ 运算的半群如上表)

Define the function $f: A^* \rightarrow A$ by

$$f(\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ has an odd number of 1's} \\ 0 & \text{if } \alpha \text{ has an even number of 1's} \end{cases}$$

It is easy to verify that if α and β are any elements of A^* , then $f(\alpha \bullet \beta) = f(\alpha) + f(\beta)$.

Thus f is a homomorphism.

Note

For both an isomorphism and a homomorphism, *the image of a product is the product of the images.*

- 同态和同构：积的像是像的积
- $f(a * b) = f(a) *' f(b)$

Theorem 3

Let

$(S, *)$ and $(T, *')$ be monoids with identities e and e' , respectively.

$f: S \rightarrow T$ be a homomorphism from $(S, *)$ *onto* $(T, *')$.

Then

$$f(e) = e'$$

Proof: similarly to the proof of theorem 2.

Theorem 4

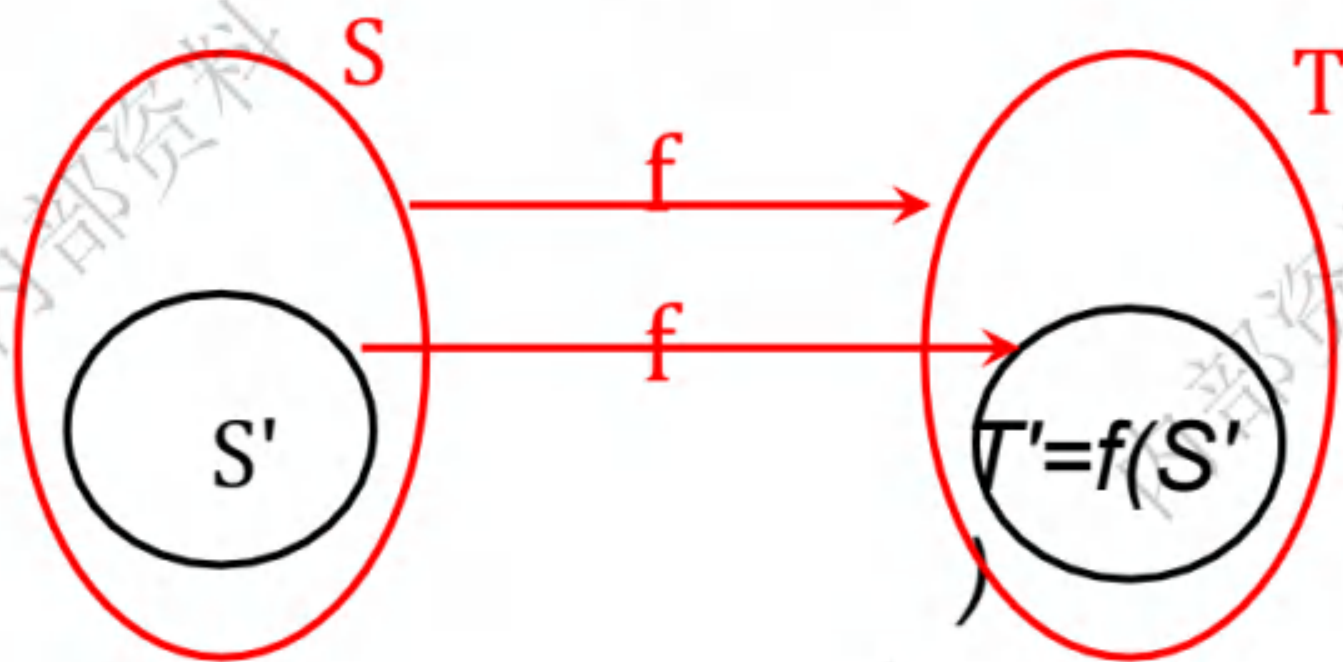
Let —— 子半群的同态 f 下的像是子半群

f be a homomorphism from a semigroup $(S, *)$ to a semigroup $(T, *')$.

If S' is a subsemigroup of $(S, *)$,

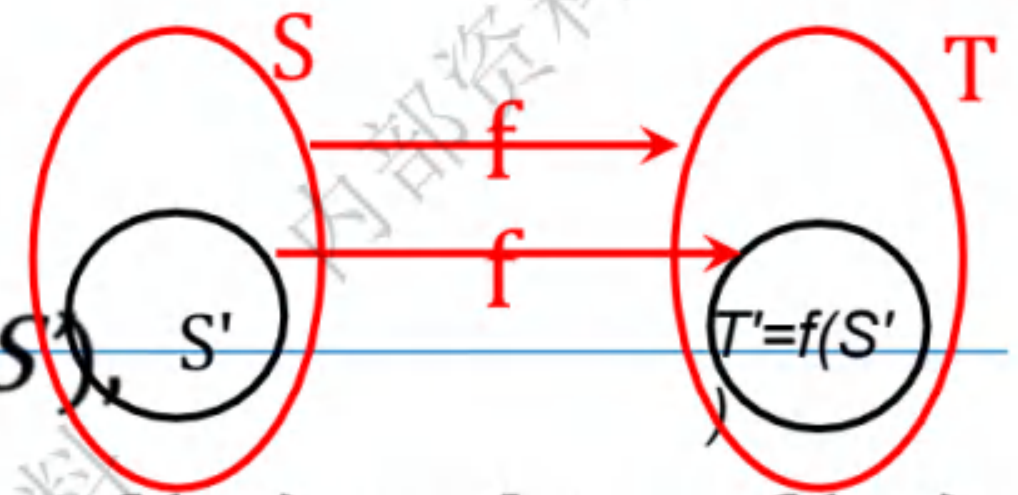
Then

$f(S') = \{t \in T \mid t = f(s) \text{ for some } s \in S'\}$, the image of S' under f , is a subsemigroup of $(T, *')$.



Theorem 4 - Proof

Let t_1 and t_2 are any elements of $f(S')$,



there exist s_1 and s_2 in S' with $t_1 = f(s_1)$ and $t_2 = f(s_2)$.

$$t_1 *' t_2 = f(s_1) *' f(s_2) = f(s_1 * s_2) = f(s_3) \dots s_3 \in S'$$

Hence $t_1 *' t_2 = f(s_3) \in f(S')$.

Thus $f(S')$ is closed under the operation $*'$.

Since the associative property holds in T, it holds in $f(S')$.

So $f(S')$ is a subsemigroup of $(T, *')$

Theorem 5

If f is a homomorphism from a commutative semigroup $(S, *)$ *onto* a semigroup $(T, *')$, then $(T, *')$ is also commutative.

Theorem 5 - Proof

$$f: S \rightarrow T$$

Let t_1 and t_2 be any elements of T . There exist s_1 and s_2 , in S such that

- $\forall t_1, t_2 \in T$
- $t_1 = f(s_1)$ and $t_2 = f(s_2)$

Therefore

- $t_1 *' t_2 = f(s_1) *' f(s_2) = f(s_1 * s_2) = f(s_2 * s_1) = f(s_2) *' f(s_1) = t_2 *' t_1$.

Hence $(T, *')$ is also commutative.

Homework

20,22,28@9.2

9.3 Products and Quotients of Semigroups

乘积半群、商半群

Products and Quotients of semigroups

- Products semigroups (积半群)
- Congruence relation (同余关系)
- Quotients of semigroups (商半群)
- Fundamental Homomorphism Theorem (基本同态原理)

Theorem 1

积半群定义

If $(S, *)$ and $(T, *')$ are semigroups, then $(S \times T, *'')$ is a semigroup, where $*''$ is defined by

$$(s_1, t_1) *'' (s_2, t_2) = (s_1 * s_2, t_1 *' t_2)$$

Proof: omitted.

What about **monoid**?

if S and T are monoids with identities e_S and e_T , respectively then $S \times T$ is a monoid with **identity** (e_S, e_T) .

积半群（续）

令： $Q = S \times T = \{(s_1, t_1), (s_1, t_2), \dots, (s_2, t_1), (s_2, t_2), \dots, (s_i, t_j), \dots\}$

- $(S \times T, *) \Leftrightarrow (Q, *)$
- 运算 $*$ 表示为 $Q \times Q \rightarrow Q$
- 任取2个序偶运算 $(s_1, t_1) * (s_2, t_2) = (s_3, t_3)$
- $s_3 = s_1 * s_2, \quad t_3 = t_1 *' t_2$
- $(s_1, t_1) * (s_2, t_2) = (s_1 * s_2, t_1 *' t_2)$

Congruence relation – 同余关系

An equivalence relation R on the semigroup $(S, *)$ is called a **congruence (同余) relation** if $a R a'$ and $b R b'$ imply $(a * b) R (a' * b')$

等价关系不一定是同余关系，因为同余必须与定义在 S 上二元运算密切相关

Example 1

Consider the **semigroup $(\mathbb{Z}, +)$** and the equivalence relation R on $\mathbb{Z}$ defined by

$a R b$ **if and only if** $a \equiv b \pmod{2}$.

If $a \equiv b \pmod{2}$, then $2 \mid (a-b)$.

R 是半群 $\mathbb{Z}$ 上的模2同余等价关系,

Show that this relation is a congruence relation。

Example 1 - Solution

R is an equivalence relation (omitted).

R is a congruence relation

If $a \equiv b \pmod{2}$ and $c \equiv d \pmod{2}$

$2 \mid a-b$ and $2 \mid c-d$.

So $a-b = 2m$ and $c-d = 2n$, where m and n are integers.

aRb , and $cRd \Rightarrow ? (a * c)R(b * d)$

$$(a-b) + (c-d) = 2m + 2n$$

$$(a+c) - (b+d) = 2(m+n),$$

so $a+c \equiv b+d \pmod{2}$.

Hence the relation is a congruence relation.

Example 1-续

Consider the **semigroup** $(\mathbb{Z}, *)$ and the equivalence relation R on $\mathbb{Z}$ defined by

$a R b$ **if and only if** $a \equiv b \pmod{2}$.

If $a \equiv b \pmod{2}$, then $2 \mid (a-b)$.

is this relation a congruence relation?

运算 ***** 表示 $\max(a, b)$

$$[0] = \{0, 2, 4, 6, 8, \dots\} \quad 2 R 4$$

$$[1] = \{1, 3, 5, 7, 9, \dots\} \quad 1 R 5$$

$$\text{Max}(2, 1) R \text{Max}(4, 5) ?$$

Example 3

Consider the semigroup $(\mathbb{Z}, +)$

$$f(x) = x^2 - x - 2.$$

Define

$a R b$ *if and only if* $f(a) = f(b)$.

Is R a congruence relation?

It is easy to verify that R is an equivalence relation, but R is not a congruence relation?

$$-1 R 2, \text{ since } f(-1) = f(2) = 0$$

$$-2 R 3, \text{ since } f(-2) = f(3) = 4$$

$$\text{but } (-1 + -2) \not R (2 + 3), \text{ since } f(-3) = 10 \text{ and } f(5) = 18$$

R 是等价关系，但不是同余关系

Note

Recall that an equivalence relation R on the semigroup $(S, *)$ determines a partition of S .
Let $[a] = R(a)$ be the equivalence class containing a .
 S/R denote the set of all equivalence classes.

Theorem 2---商半群

Let R be a **congruence relation** on the semigroup $(S, *)$.

Consider the relation $\odot$ from $S/R \times S/R$ to S/R in which the ordered pair $([a], [b])$ is, for a and b in S , related to $[a * b]$. (构造商半群 **$(S/R, \odot)$**)

- $\odot([a], [b]) = [a] \odot [b] = [a * b]$, is a function from $S/R \times S/R$ to S/R
- **$(S/R, \odot)$** is a semigroup. S/R is called the **quotient semigroup**(商半群)

R 是半群 $(S, *)$ 上的同余关系

$\odot$ 是 $S/R \times S/R$ to S/R 的函数

$(S/R, \odot)$

1. $S = \{a, b, c \dots a', b', c' \dots\}$

2. $S/R = \{[a], [b], [c] \dots\}$

3. 假定等价类 $[a] = \{a, a', a'' \dots\}$, $[b] = \{b, b', b'' \dots\}$
 $\Rightarrow a R a', b R b'$

4. $S/R = \{[a], [b], [c], \dots\}$

5. $S/R \times S/R = \{([a], [a]), ([a], [b]), ([a], [c]) \dots, ([b], [a]), ([b], [b]), \dots\}$

6. $f: S/R \times S/R \rightarrow S/R$

7. $f([a], [b]) = [c] \Leftrightarrow [a] \odot [b] = [c]$

Proof

- $\odot$ is a binary operation (证二元运算封闭)

Suppose that $([a], [b]) = ([a'], [b'])$.

$a R a'$ and $b R b'$,

$a * b R a' * b'$, since R is a congruence relation.

Thus $[a * b] = [a' * b']$;

($[a] \odot [b] = [a * b] = [c]$ — 因为 $a * b = c$ 在原代数系统中运算)

$\odot$ is a function, is a binary operation on S/R .

- $\odot$ is an associative (证可结合)

$$[a] \odot ([b] \odot [c]) = [a] \odot [b * c] = [a * (b * c)] = [(a * b) * c] = [a * b] \odot [c] = ([a] \odot [b]) \odot [c].$$

Corollary

Let R be a congruence relation on the **monoid** $(S, *)$. If we define the operation $\odot$ in S/R by $[a] \odot [b] = [a * b]$, then $(S/R, \odot)$ is a monoid.

么半群导出的商集也是么半群

Proof

If e is the identity in $(S, *)$, then it is easy to verify that $[e]$ is the identity in $(S/R, \odot)$.

$$[a] \odot [e] = [a * e] = [a] = [e * a] = [e] \odot [a]$$

Example 5

$\oplus$	[0]	[1]	[2]	[3]
[0]	[0]	[1]	[2]	[3]
[1]	[1]	[2]	[3]	[0]
[2]	[2]	[3]	[0]	[1]
[3]	[3]	[0]	[1]	[2]

$(\mathbf{Z}, +)$

$a R b$ if and only if $a = b \pmod{n}$.

R is an equivalence relation

$\equiv \pmod{4}$ is a congruence relation

$$[0] = \{\dots, -8, -4, 0, 4, 8, 12, \dots\} = [4] = [8] = \dots$$

$$[1] = \{\dots, -7, -3, 1, 5, 9, 13, \dots\} = [5] = [9] = \dots$$

$$[2] = \{\dots, -6, -2, 2, 6, 10, 14, \dots\} = [6] = [10] = \dots$$

$$[3] = \{\dots, -5, -1, 3, 7, 11, 15, \dots\} = [7] = [11] = \dots$$

$\mathbf{Z}/\equiv \pmod{4}$ or $\mathbf{Z}_4$ is a monoid with identity [0] and operation $[a] \oplus [b] = [a+b]$.

Theorem 3-自然同态

Let

R be a congruence relation on a semigroup $(S, *)$,
and
 $(S/R, \odot)$ be the corresponding
quotient semigroup.

Then the function

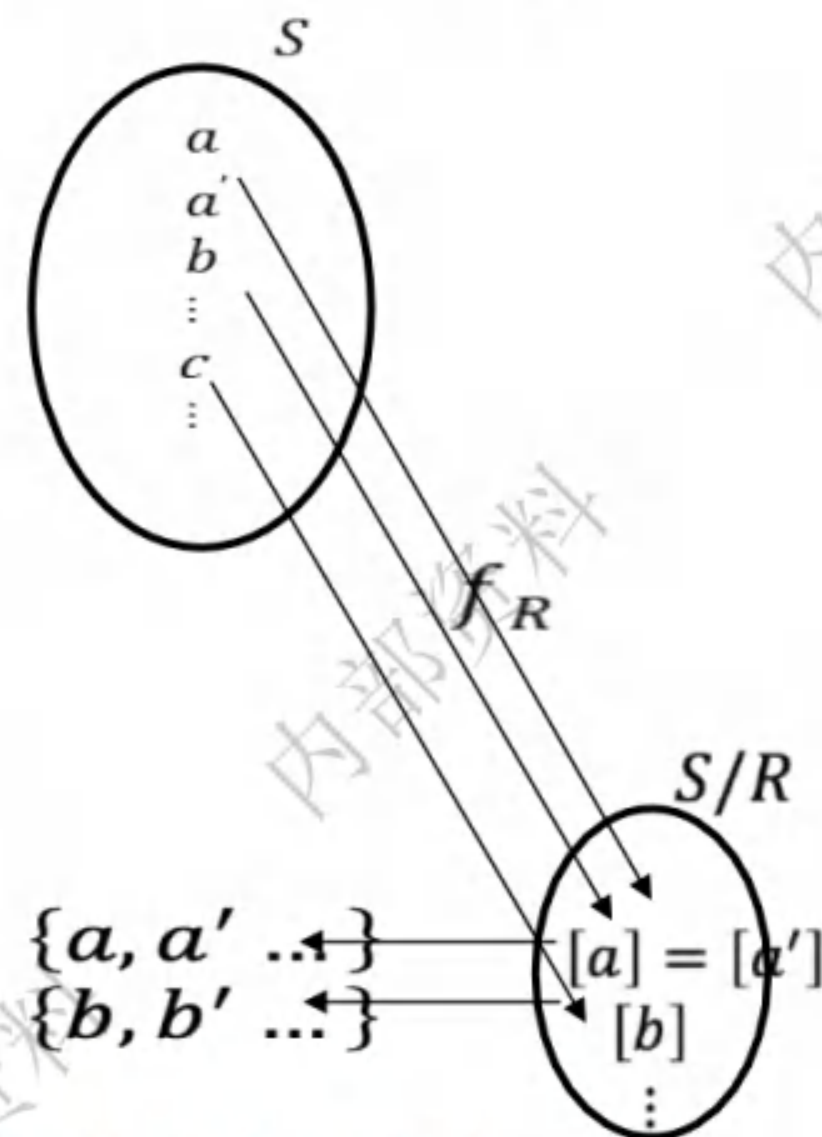
$$f_R: S \rightarrow S/R$$

defined by

$$f_R(a) = [a]$$

is an **onto** homomorphism,

called the **natural homomorphism**(自然同态).



R 同余，满射的 f_R 是 $S \rightarrow S/R$ 的自然同态

Theorem 3 - Proof

If $\forall [a] \in S/R$, then

$$f_R(a)=[a],$$

so f_R is an onto function.

if a and b are elements of S ,

($\forall a, b \in S$), then

$$f_R(a * b)=[a * b]=[a] \odot [b]$$

$$f_R(a) \odot f_R(b)=[a] \odot [b]$$

Hence $[a] \odot [b]=[a * b]$

so f_R is a homomorphism.



Fundamental Homomorphism

Theorem 4— 同态基本定理

Let

$f: S \rightarrow T$ be a **homomorphism** of the semigroup $(S, *)$ **onto** the semigroup $(T, *')$.

R be the relation on S defined by

$a R b$ **if and only if** $f(a) = f(b)$, for a and b in S .

Then

(1) R is a congruence relation.



Proof

R is an equivalence relation(证等价关系)

$a R a$ for every $a \in S$, since $f(a) = f(a)$.

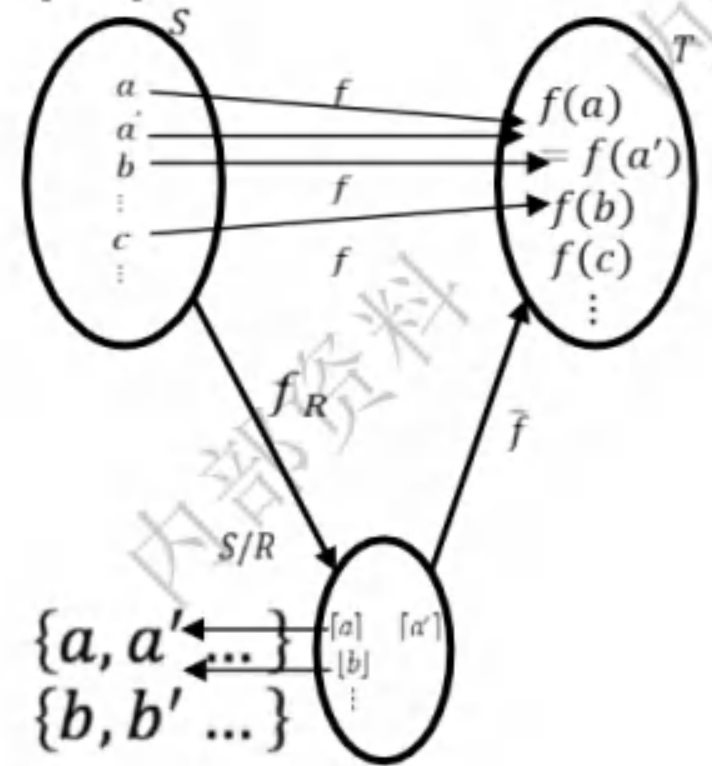
if $a R b$, then $f(a) = f(b)$, so $b R a$.

if $a R b$ and $b R c$,

$f(a) = f(b)$ and $f(b) = f(c)$,

so $f(a) = f(c)$ and $a R c$.

Hence R is an equivalence relation.



Proof

R is a congruence relation. (证同余关系)

Suppose that $a R a_1$ and $b R b_1$.

$f(a) = f(a_1)$ and $f(b) = f(b_1)$.

因为 f 是同态

$f(a * b) = f(a) *' f(b) = f(a_1) *' f(b_1) = f(a_1 * b_1)$, since f is a homomorphism,

Hence $(a * b) R (a_1 * b_1)$

Proof-证同构

Define a relation $\bar{f} = \{([a], f(a)) \mid [a] \in S/R\}$ from S/R to T , then

$\bar{f}$ is function. (证是函数)

Suppose that $[a] = [a']$.

$a R a'$, so $f(a) = f(a')$, which implies that $\bar{f}$ is a function.

write $\bar{f}: S/R \rightarrow T$, where $\bar{f}([a]) = f(a)$ for $[a] \in S/R$.

$\bar{f}$ is one to one. (证是单射)

Suppose that $\bar{f}([a]) = \bar{f}([a'])$.

$f(a) = f(a')$, so $a R a'$, which implies that $[a] = [a']$.

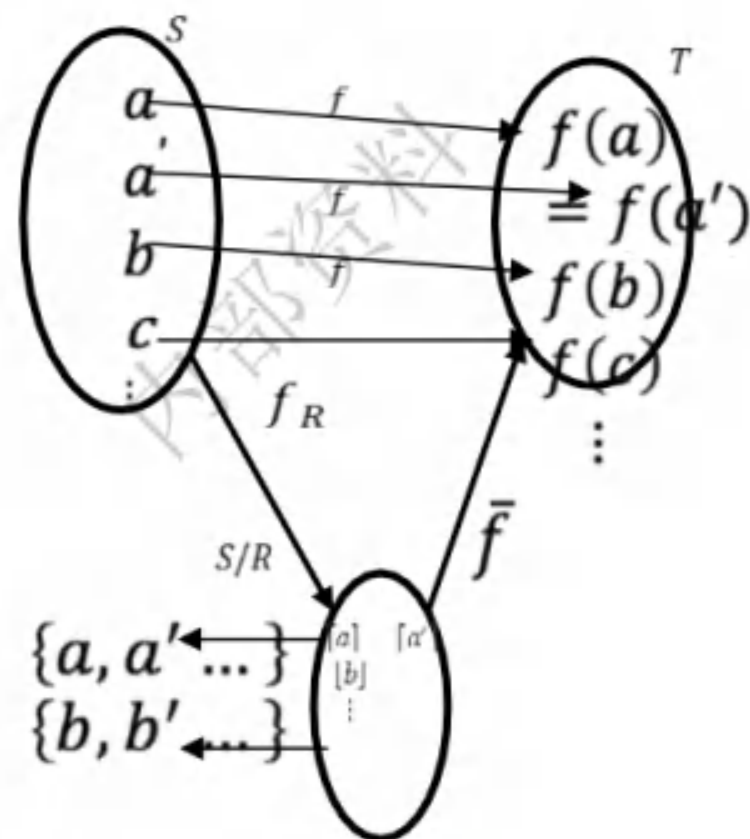
Hence $\bar{f}$ is one to one.

$\bar{f}$ is onto. (证是满射)

Suppose that $\forall t \in T$.

$f(a) = t$ for some element a in S ($\exists a \in S$), since f is onto.

$\bar{f}([a]) = f(a) = t$, so $\bar{f}$ is onto.



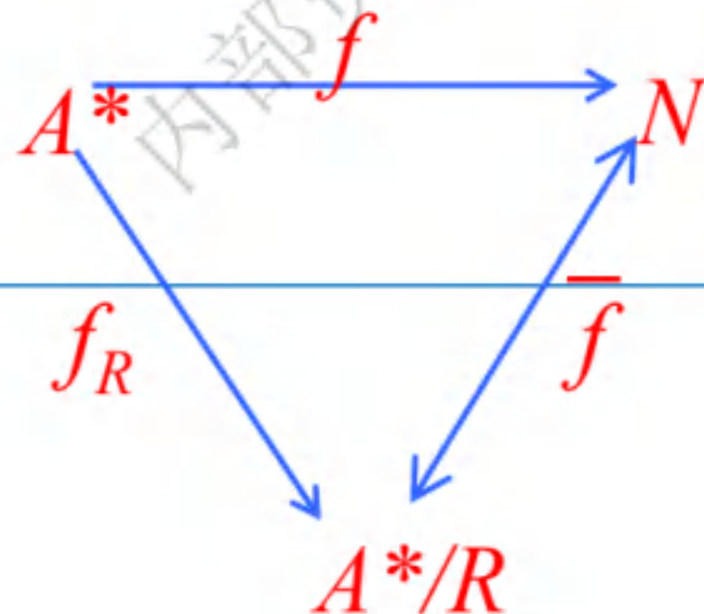
Proof

$\bar{f}$ is an isomorphism. (证同构)

$$\begin{aligned}\bar{f}([a] \odot [b]) \\ &= \bar{f}([a * b]) \\ &= f(a * b)\end{aligned}$$

$$\begin{aligned}\bar{f}([a]) *' \bar{f}([b]). \\ &= f(a) *' f(b) \text{ (因为 } S, \tau \text{ 同态)} \\ &= f(a * b)\end{aligned}$$

Example 6



$$A = \{0, 1\}$$

$$(A^*, \cdot) \quad (N, +)$$

$f: A^* \rightarrow N$, 定义为

$f(\alpha)$ = 等于 “ α 中含1的个数”

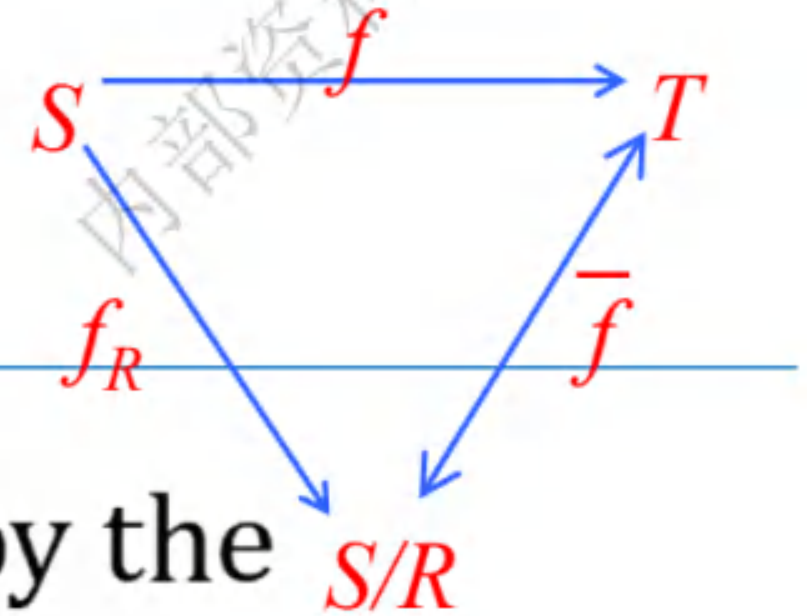
$R: \alpha R \beta$ if and only if $f(\alpha) = f(\beta)$, 可验证同态,

$A^* / R \cong N$ $f: A^* / R \rightarrow N$ (定理4可知同构)

$$f_R(\alpha) = [\alpha]$$

$$\bar{f}([\alpha]) = f(\alpha)$$

Note



Theorem 4(b) can be described by the diagram above

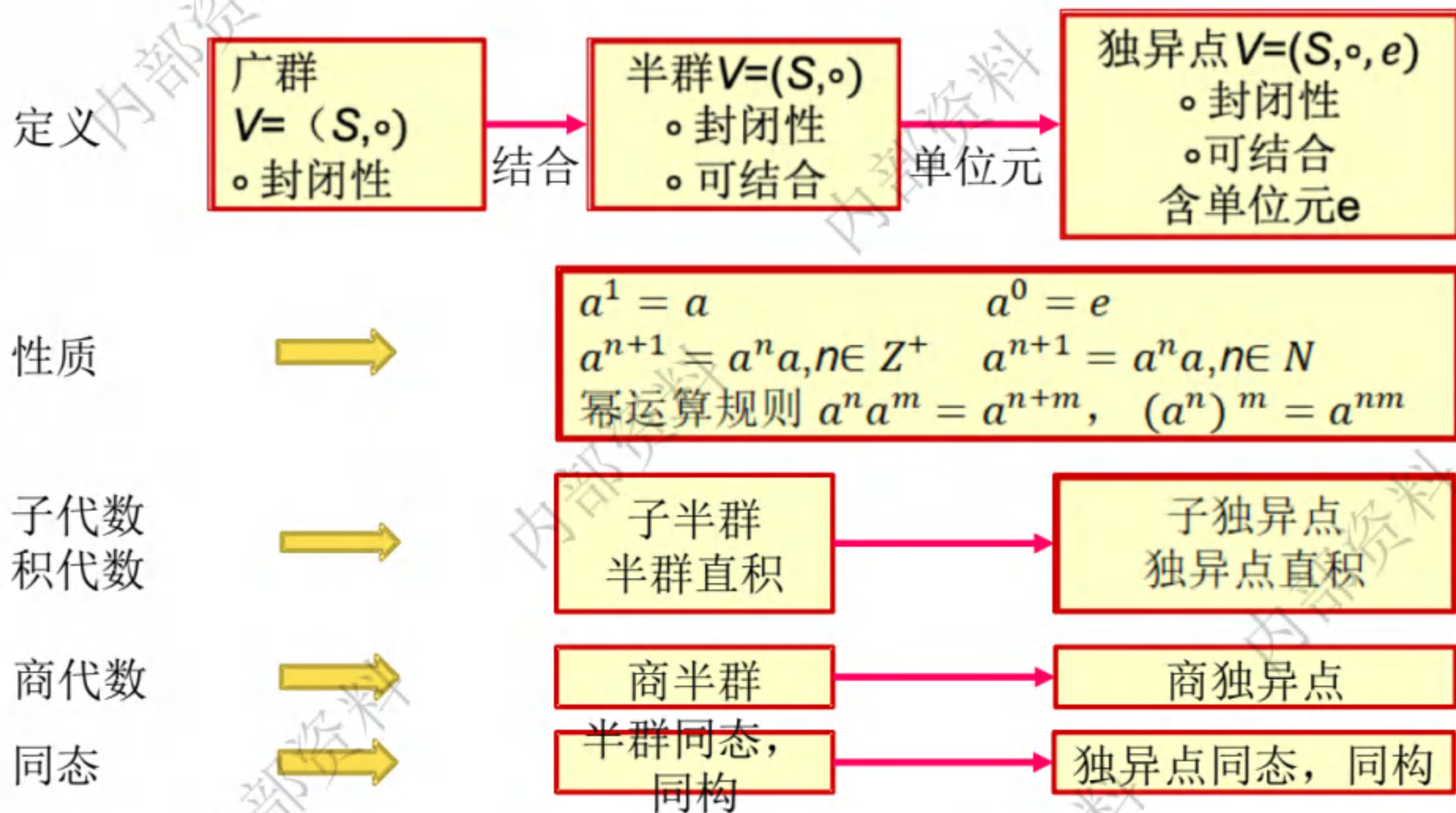
f_R is the natural homomorphism.

It follows from the definitions of f_R and f that

$$\bar{f} \circ f_R = f$$

Since

$$(\bar{f} \circ f_R)(a) = \bar{f}(f_R(a)) = \bar{f}([a]) = f(a).$$



Homework

22, 26, 32@9.2

4, 12, 22, 24, 26@9.3

Determine whether the relation R on the semigroup S is a congruence relation:

- ☐ A $S=\mathbb{Z}$ under the operation of ordinary addition; $a R b$ if only if 2 does not divide $a-b$.
- ☒ B $S=\mathbb{Z}$ under the operation of ordinary addition; $a R b$ if only if $a+b$ is even.
- ☐ C S = any semigroup; $a R b$ if only if $a=b$.
- ☒ D S =the set of all rational numbers under the operation of multiplication; $a/b R c/d$ if and only if $ad=bc$
- ☒ E $S=\mathbb{Z}$ under the operation of ordinary addition; $a R b$ if and only if $a \equiv b \pmod{3}$.



Thanks for your attention!

Coming up next ...

Groups

